How do Bronze tables relate to raw data in a data lakehouse architecture?

A. Bronze tables contain raw data that has been aggregated, unlike the unaggregated raw data.

B. Bronze tables hold a more refined view of data compared to raw data.

C. Bronze tables are where raw data is stored with a schema applied to it.

D. Bronze tables have less data volume than the original raw data files.

E. Bronze tables provide more accurate data than the raw data they originate from.

A data engineering team is tasked with optimizing both the cost and performance of their data pipeline in a cloud environment. They are considering implementing incremental processing to achieve these goals. The team is particularly concerned with minimizing costs associated with data processing while ensuring that the pipeline can scale to handle increasing volumes of data without significant latency. Given this scenario, which of the following best describes incremental processing and its implementation to meet the team's objectives? (Choose two options that best apply.)

A. Incremental processing involves processing the entire dataset from scratch in each run to ensure data accuracy, which may increase costs but guarantees up-to-date information.

B. Incremental processing is a technique where only the new or changed data since the last processing run is processed, significantly reducing the amount of data handled and thus lowering costs and improving performance.

C. Incremental processing requires the use of real-time data streaming technologies to process data as it arrives, which may not always be cost-effective for all types of data pipelines.

D. Incremental processing can be implemented by tracking changes in the data source, such as using timestamps or change data capture (CDC), to identify and process only the delta, optimizing both cost and performance.

31. A pipeline adds a calculated column to a raw table and filters out invalid records. What 'hop' occurred?

A Silver to Gold

B Bronze to Silver

C Raw to Bronze

D Bronze to Gold

52. A column 'json_payload' contains JSON strings. Which function converts these strings into a Spark Struct type?

A to_json()

B get_json_object()

C from_json()

D explode()

To ensure future management of table permissions by any member of a group rather than a single individual, which command should you use to transfer the ownership of tables from a departing team member to a group?

A. TRANSFER OWNER table_name to 'group'

B. GRANT OWNER On table_name to 'group'

C. ALTER TABLE table_name OWNER to 'group'

D. GRANT OWNER table_name to 'group'

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9. A data engineer needs to create an empty table with a specific schema. Which command will succeed regardless of whether a table with that name already exists?

A. INSERT INTO table_name

B. MERGE INTO table_name

C. CREATE OR REPLACE TABLE table_name

D. DROP TABLE table_name

Which of the following describes when to use the CREATE STREAMING LIVE TABLE (formerly CREATE INCREMENTAL LIVE TABLE) syntax over the CREATE LIVE TABLE syntax when creating Delta Live Tables (DLT) tables using SQL?

A. CREATE STREAMING LIVE TABLE should be used when the subsequent step in the DLT pipeline is static.

B. CREATE STREAMING LIVE TABLE should be used when data needs to be processed incrementally.

C. CREATE STREAMING LIVE TABLE is redundant for DLT and it does not need to be used.

D. CREATE STREAMING LIVE TABLE should be used when data needs to be processed through complicated aggregations.

E. CREATE STREAMING LIVE TABLE should be used when the previous step in the DLT pipeline is static.

A data engineer has developed a data pipeline to ingest data from a JSON source using Auto Loader, but the engineer has not provided any type inference or schema hints in their pipeline. Upon reviewing the data, the data engineer has noticed that all of the columns in the target table are of the string type despite some of the fields only including float or boolean values.
Which of the following describes why Auto Loader inferred all of the columns to be of the string type?

A	There was a type mismatch between the specific schema and the inferred schema
B	JSON data is a text-based format
C	Auto Loader only works with string data
D	All of the fields had at least one null value
E	Auto Loader cannot infer the schema of ingested data

Question 43

A data engineering manager has noticed that each of the queries in a Databricks SQL dashboard takes a few minutes to update when they manually click the “Refresh” button. They are curious why this might be occurring, so a team member provides a variety of reasons on why the delay might be occurring. Which of the following reasons fails to explain why the dashboard might be taking a few minutes to update?

A	The SQL endpoint being used by each of the queries might need a few minutes to start up.
B	The queries attached to the dashboard might take a few minutes to run under normal circumstances.
C	The queries attached to the dashboard might first be checking to determine if new data is available.
D	The Job associated with updating the dashboard might be using a non-pooled endpoint.
E	The queries attached to the dashboard might all be connected to their own, unstarted Databricks clusters.

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A data engineer wants to set a daily refresh schedule for their Databricks SQL dashboard and wishes to have the associated SQL endpoint active only when needed. Which method should the data engineer employ to keep the SQL endpoint's operational time to a minimum for the dashboard's refresh schedule?

A	Configure the dashboard's SQL endpoint to operate in a serverless mode.
B	Make sure the dashboard's SQL endpoint is not linked to any of the queries' SQL endpoints.
C	Ensure the dashboard's SQL endpoint aligns with each query's SQL endpoint.
D	Decrease the cluster size of the SQL endpoint.
E	Activate the Auto Stop feature for the SQL endpoint.

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56. Which object allows saving a complex SQL query for reuse across sessions without physically storing the result data?

A Delta Table

B Common Table Expression (CTE)

C View

D Temporary View

Suppose you have a DataFrame `df` with a column `url` containing URLs. How would you extract the domain name from these URLs and create a new column `domain` using Spark? Provide the code snippet.

A. df.withColumn('domain', regexp_extract(col('url'), 'https?://([^/]+)', 1))

B. df.select(regexp_extract(col('url'), 'https?://([^/]+)', 1).alias('domain'))

C. df.withColumn('domain', split(col('url'), '/')[2])

D. df.select(split(col('url'), '/')[2].alias('domain'))

Which of the following data workloads will utilize a Gold table as its source?

A. A job that enriches data by parsing its timestamps into a human-readable format

B. A job that aggregates uncleaned data to create standard summary statistics

C. A job that cleans data by removing malformed records

D. A job that queries aggregated data designed to feed into a dashboard

E. A job that ingests raw data from a streaming source into the Lakehouse

A data engineer needs to create a table in Databricks using data from a CSV file at location `/path/to/csv`. They run the following command:

```
CREATE TABLE new_table
_________
OPTIONS (
    header = "true",
    delimiter = "|"
)
LOCATION "path/to/csv"
```

Which of the following lines of code fills in the above blank to successfully complete the task?

A. None of these lines of code are needed to successfully complete the task

B. USING CSV

C. FROM CSV

D. USING DELTA

E. FROM "path/to/csv"

Which of the following describes the storage organization of a Delta table?

A. Delta tables are stored in a single file that contains data, history, metadata, and other attributes.

B. Delta tables store their data in a single file and all metadata in a collection of files in a separate location.

C. Delta tables are stored in a collection of files that contain data, history, metadata, and other attributes.

D. Delta tables are stored in a collection of files that contain only the data stored within the table.

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What is used by Spark to record the offset range of the data being processed in each trigger in order for Structured Streaming to reliably track the exact progress of the processing so that it can handle any kind of failure by restarting and/or reprocessing?

A. Checkpointing and Write-ahead Logs

B. Replayable Sources and Idempotent Sinks

C. Write-ahead Logs and Idempotent Sinks

D. Checkpointing and Idempotent Sinks

A data engineer is responsible for executing a daily operation that involves copying sales data from the previous day into a table named `transactions`. The sales data for each day is stored in separate files located in the directory "/transactions/raw". To perform today's data copying task, the data engineer runs a specific command. However, after executing the command, the data engineer observes that the number of records in the `transactions` table remains unchanged. Which of the following could explain why the command did not add any new records to the table?

A. The format of the files to be copied were not included with the FORMAT_OPTIONS keyword.

B. The names of the files to be copied were not included with the FILES keyword.

C. The previous day's file has already been copied into the table.

D. The PARQUET file format does not support COPY INTO.

E. The COPY INTO statement requires the table to be refreshed to view the copied rows.

Where is the Databricks web application hosted according to the Databricks Lakehouse architecture?

A. Databricks Filesystem

B. Control plane

C. Customer Cloud Account

D. Data plane

E. Databricks-managed cluster

Which statement about Auto Loader's exactly-once guarantee is correct?

A. It is achieved by watermarking late data.

B. It is achieved by storing file metadata in RocksDB at the checkpoint location.

C. It is achieved by file notification mode only.

D. It is achieved by schema evolution.

The operations team wants to monitor a recently launched product by setting up an email alert that triggers when the number of units sold increases by more than 10,000 units, with checks every 5 minutes. Complete the steps needed to achieve this by filling in the blanks:
· Create a ___ query that calculates total units sold · Setup a ____ with the query on the trigger condition 'Units Sold > 10,000' · Setup a ____ to run every 5 minutes · Add a destination _____

A Python, Job, SQL Cluster, email address

B SQL, Alert, Refresh, email address

C SQL, Job, SQL Cluster, email address

D SQL, Job, Refresh, email address

E Python, Job, Refresh, email address

A data engineer wants to create a data entity from a couple of tables. The data entity must be used by other data engineers in other sessions. It also must be saved to a physical location. Which of the following data entities should the data engineer create?

A. Table

B. Function

C. View

D. Temporary view

A data engineer and data analyst are working together on a data pipeline. The data engineer is working on the raw, bronze, and silver layers of the pipeline using Python, and the data analyst is working on the gold layer of the pipeline using SQL. The raw source of the pipeline is a streaming input. They now want to migrate their pipeline to use Delta Live Tables. Which of the following changes will need to be made to the pipeline when migrating to Delta Live Tables?

A. None of these changes will need to be made

B. The pipeline will need to stop using the medallion-based multi-hop architecture

C. The pipeline will need to be written entirely in SQL

D. The pipeline will need to use a batch source in place of a streaming source

E. The pipeline will need to be written entirely in Python

Given the following two tables, which JOIN operation should be used in the blank to ensure the query returns all students, including those not enrolled in any course, with their course_id if available?

```
SELECT students.name, students.age, enrollments.course_id
FROM students
_____________ enrollments
ON students.student_id = enrollments.student_id
```

A RIGHT JOIN

B INNER JOIN

C LEFT JOIN

D ANTI JOIN

E CROSS JOIN

A data analyst has a series of queries in a SQL program. The data analyst wants this program to run every day. They only want the final query in the program to run on Sundays. They ask for help from the data engineering team to complete this task. Which of the following approaches could be used by the data engineering team to complete this task?

A. They could submit a feature request with Databricks to add this functionality.

B. They could wrap the queries using PySpark and use Python's control flow system to determine when to run the final query.

C. They could only run the entire program on Sundays.

D. They could automatically restrict access to the source table in the final query so that it is only accessible on Sundays.

E. They could redesign the data model to separate the data used in the final query into a new table.